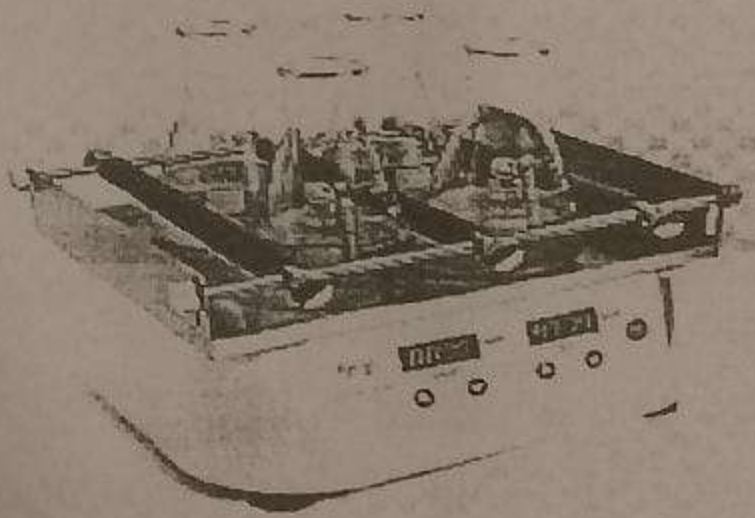
## 1. CENTRIFUGE MACHINE:

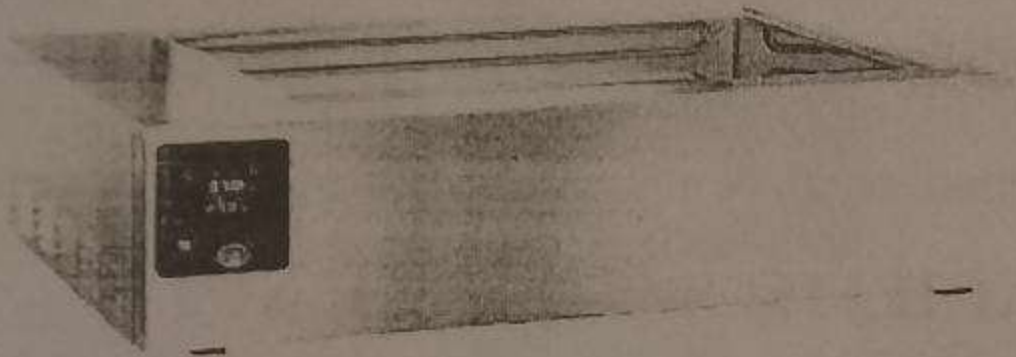


the application of centrifugal

## 3. SHAKER /MIXER:



# THERMOSTATIC BATH /WATER BATH:



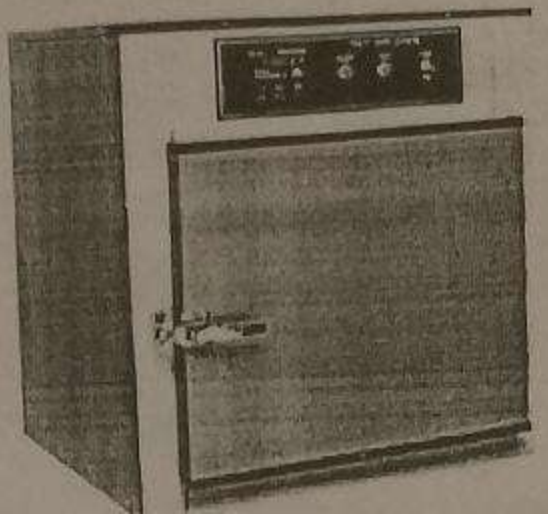
## 4. SPECTROPHOTOMETER:



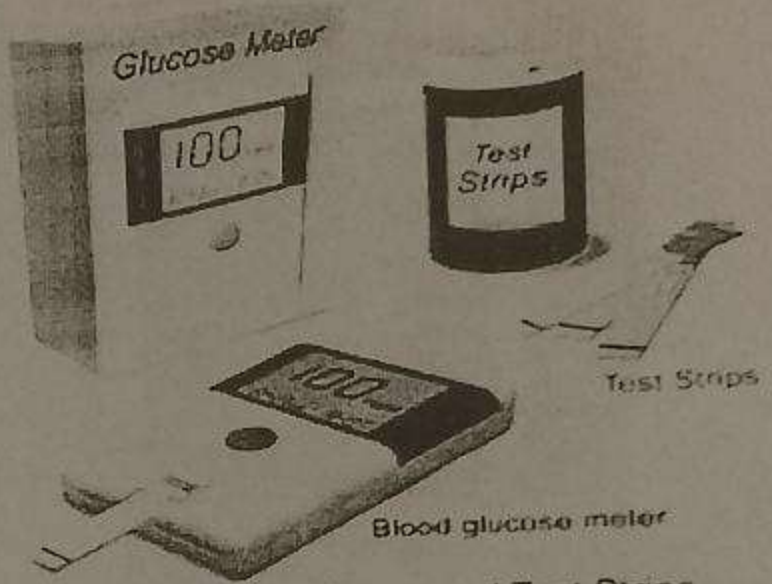
# ELECTRONIC BALANCE/ANALYTICAL BALANCE:



6. OVEN:

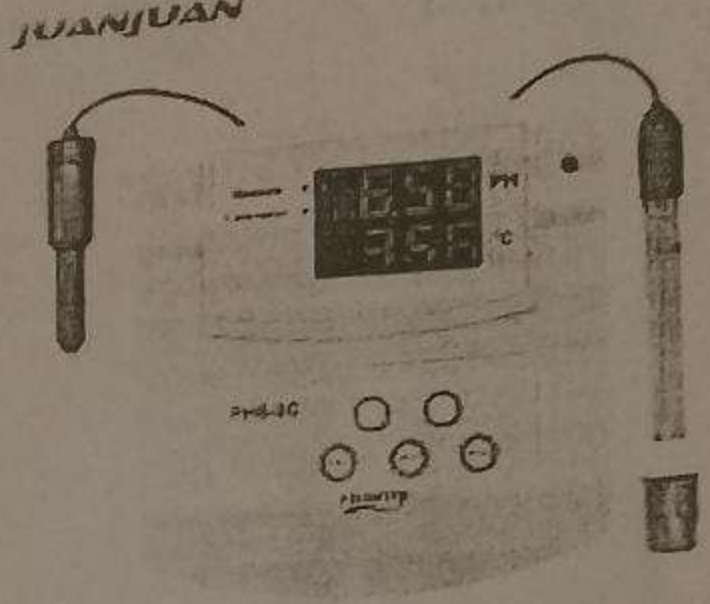


## 7. GLUCOMETER:




Blood Glucose Meter and Test Strips

8. pH METER:



# URACIL VERSUS THYMINE

| URACIL | THYMINE |
| --- | --- |
| Uracil is a pyrimidine base that is a component of RNA | Thymine is a pyrimidine base that is a component of DNA |
| Only occurs in RNA | Only occurs in DNA |
| Heterocyclic aromatic ring contains two keto groups at its C-2 and C-4 | Heterocyclic aromatic ring contains two keto groups at C-2 and C-4, and a methyl group at C-5 |
| Molar mass is 112.0868 g/mol | Molar mass is 126.1133 g/mol |
| Uridine nucleotides serve as allosteric regulators and coenzymes in plants and humans | Thymine can be derived by the methylation of uracil at its C-5 |

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# Hazard Symbols



General Warning



Biohazard



Explosive Hazard



Harmful Irritant



Poison/Toxic Material



Toxic Gas



Noise Sign



Corosive Material Hazard



High Voltage



Electrical Hazard



Laser Beam Hazard



Low Temperature Warning Symbol




H19A-S




H22A-S




H20A-S




H18A-S




H21A-S




H15A-S




H13A-S




H16A-S




H14A-S




H17A-S

- AA sequence: Rich in glycine & proline
  - Smallest AA found every 3rd position in polypeptide chain
  - facilitate formation of helical conformation
- Glycine residues → part of repeating sequence Gly X Y
- Hydroxyproline Hydroxyline → result from hydroxylation of proline & lysin. Post translational modification.
- Glycosylation

| FIBRIL FORMING COLLAGEN | NETWORK FORMING | FIBRIL ASSOCIATED |
|---|---|---|
| I → Skin, bone, tendon Blood vessel, cornea | IV → Basement membrane | IX → Cartilage |
| II → Cartilages, intervertebral disc, vitreous body | VII → Corneal & vascular endothelium | XII → Tendon, ligaments, some other tissue |
| III → Blood vessels, skin, muscle | | |

2:04 96%

AI Overview +1

Collagen is a protein that provides structure and strength to connective tissues in the body. There are many types of collagen, but the most common are types I, II, III, and IV.

## Types of collagen

- **Type I**: The most common type of collagen, found in bones, tendons, and skin
- **Type II**: Found in cartilage, which provides support for joints
- **Type III**: Found in skin, blood vessels, and muscles
- **Type IV**: Found in the layers of skin
- **Type V**: Found in the cornea, some layers of skin, hair, and placental tissue

## Collagen functions

- Gives structure, strength, and elasticity to skin
- Helps blood clot
- Cushions connective tissues

## Collagen sources

- The body produces collagen naturally
- Collagen can be found in some foods, such as fish and chicken skin

2:24 92%
google.com/search?
Google
what is antioxidant
All Images Videos Short videos Web
AI Overview +7
Antioxidants are molecules that protect cells from damage by neutralizing free radicals.
Free radicals are unstable molecules that can harm cells and DNA.
Examples of antioxidants
Show more
Antioxidant
Overview
Antioxidants are compounds that inhibit oxidation, a chemical reaction that can produce free radicals. Autoxidation leads to degradation of organic compounds, including liv... Wikipedia
People also ask
What are antioxidants?

| | | |
|---|---|---|
| Cu-transporting ATPase affected | ATP7A | ATP7B |

**Figure 29.6**
Comparison of Menkes syndrome and Wilson disease. Cu = copper; AR = autosomal recessive.



**Figure 29.7**
Kaiser–Fleischer rings.

# Blood Clotting

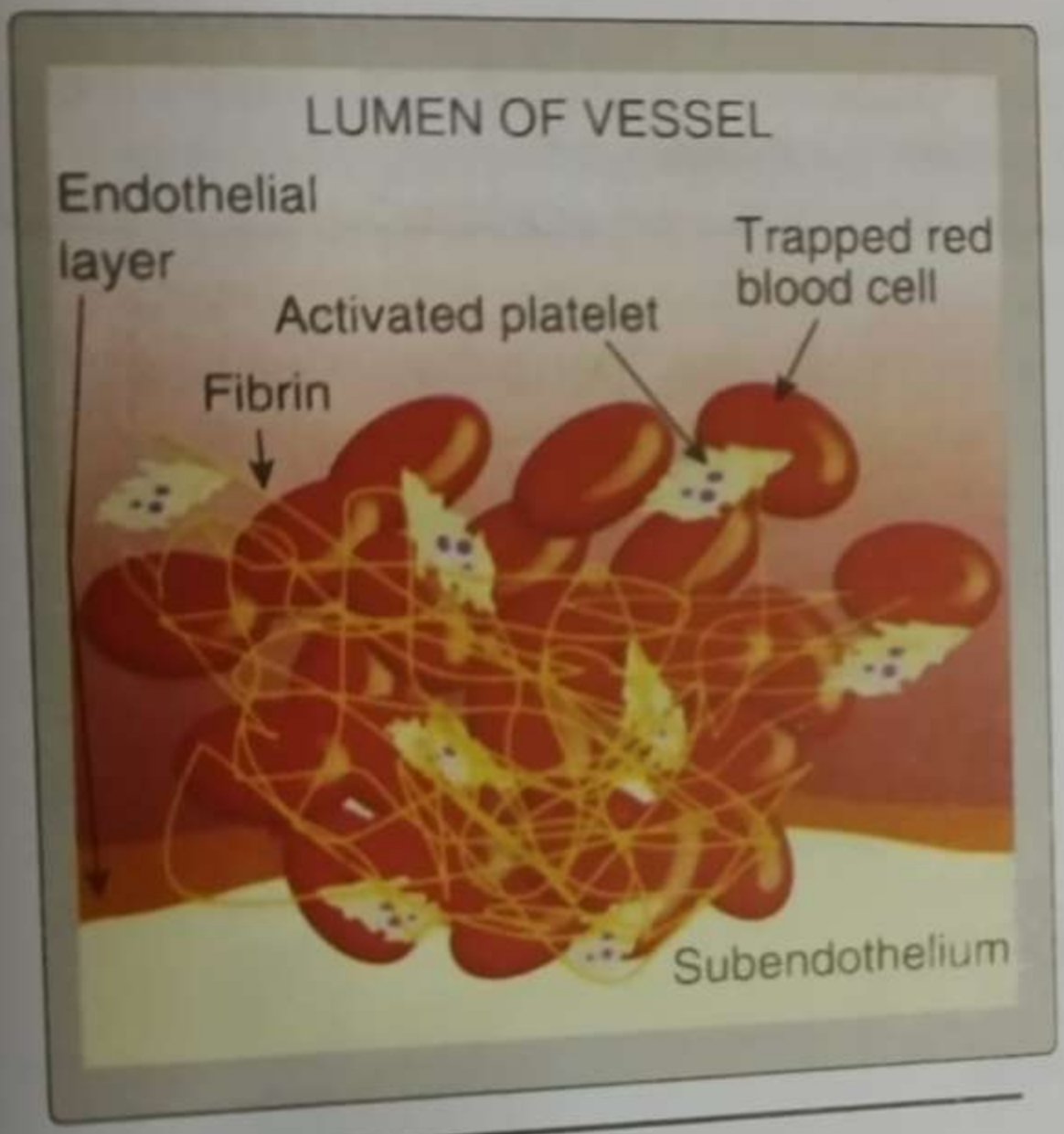



**Figure 35.1**
A blood clot formed by a plug of activated platelets and a meshwork of fibrin at the site of vessel injury.

## I. OVERVIEW

Blood clotting (coagulat
a damaged blood vesse
(hemostasis). Coagulati
and the formation of a c
lets (primary hemostasi
ondary hemostasis) that
association with membr
blood vessels (Fig. 35.1)
such that the lumen is c
tion known as thrombos
occur. This is what happ
tion [MI].) Processes to
remove the clot once ves
in hemostasis. (Note: Sep
let plug and the fibrin m
step, multicomponent p
maintain hemostasis.)

## II. SECONDARY HE FORMATION

The formation of the fibri
and involves two unique
converge to form a c

plasma protein **cerulo-**
al individuals, transferrin
$^{3+}$. Ferroportin, the only
ood in humans, is regu-
t induces internalization
n. Therefore, hepcidin is
. (Note: Transcription of
ient.)

old and/or damaged red
at is sent out of the cells
**in**, and transported by Tf
neets approximately 90%
ly for erythropoiesis.

cytes and macrophages
ts and other Fe-requiring
ted endocytosis. The $Fe^{3+}$
n ferritin), and the TfR (and
e receptor-mediated endo-
tein particles (see p. 257).
he messenger RNA for fer-
oteins and iron-responsive

ult in a microcytic, hypo-
st common anemia in the
ased hemoglobin synthe-
RBC size. Treatment is the
s depending on the sever-

accidental ingestion. (Note:
common cause of poison-
s [UL = 40 mg/day for chil-
ent is use of an Fe chelator.
etic defects. An example is
an AR disorder of Fe over-
thern European ancestry. It is
ns to the HFE (high Fe) gene.
emia ("bronze diabetes") and
ge site for Fe), pancreas, and
Fe and Tf saturation are ele-
r use of Fe chelators. (Note:
s to proteins of Fe metabolism
evels of hepcidin. It can result
of hemosiderin, an intracellular,

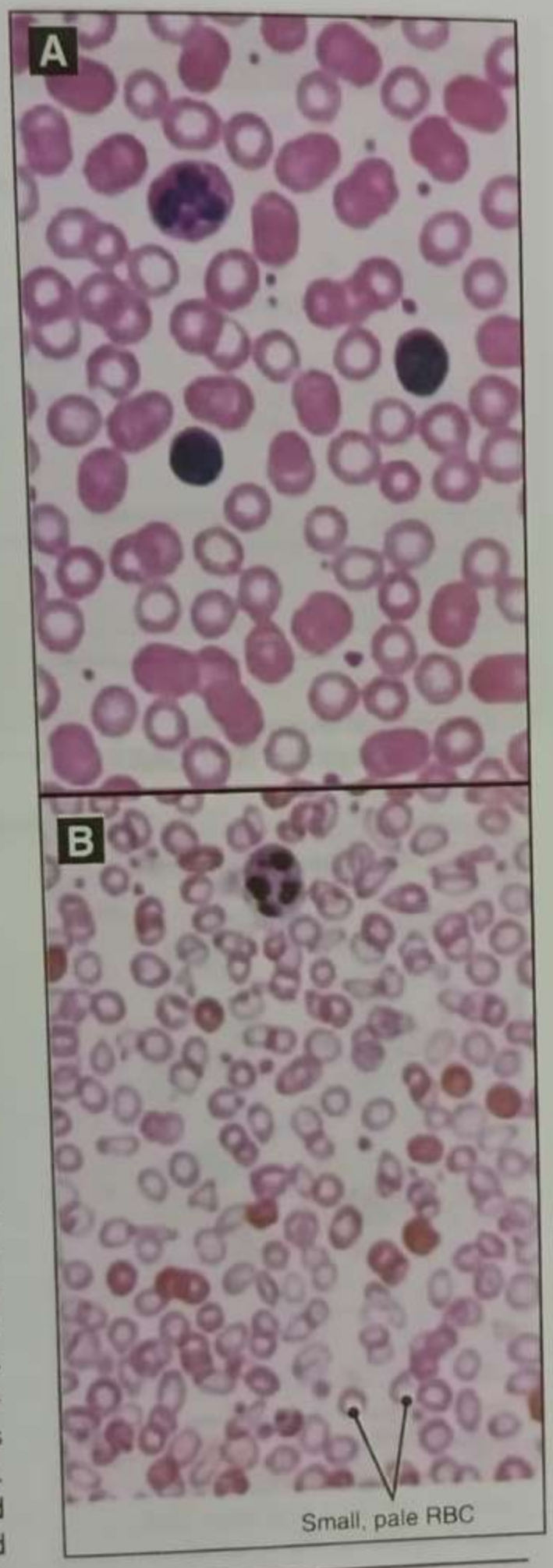



**Figure 29.9**
**A:** Normal red blood cells (RBCs).
**B:** Small (microcytic), pale (hypochromic) RBC in microcytic anemia.

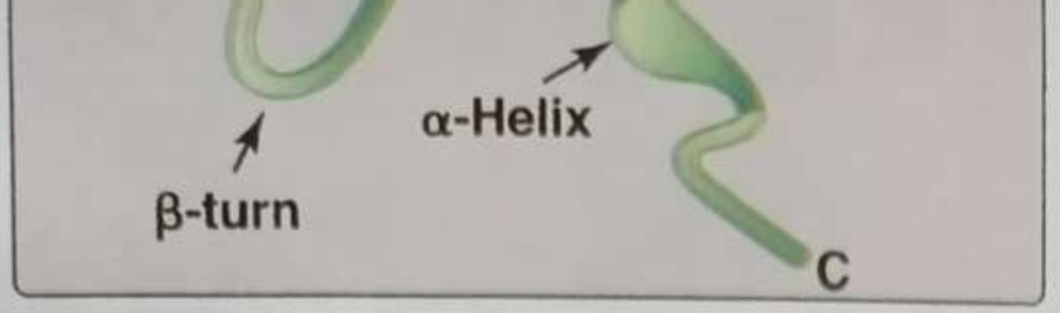



**Figure 29.11**
Zinc (Zn) finger is a common motif in proteins that bind DNA. Cys = cysteine; His = histidine.

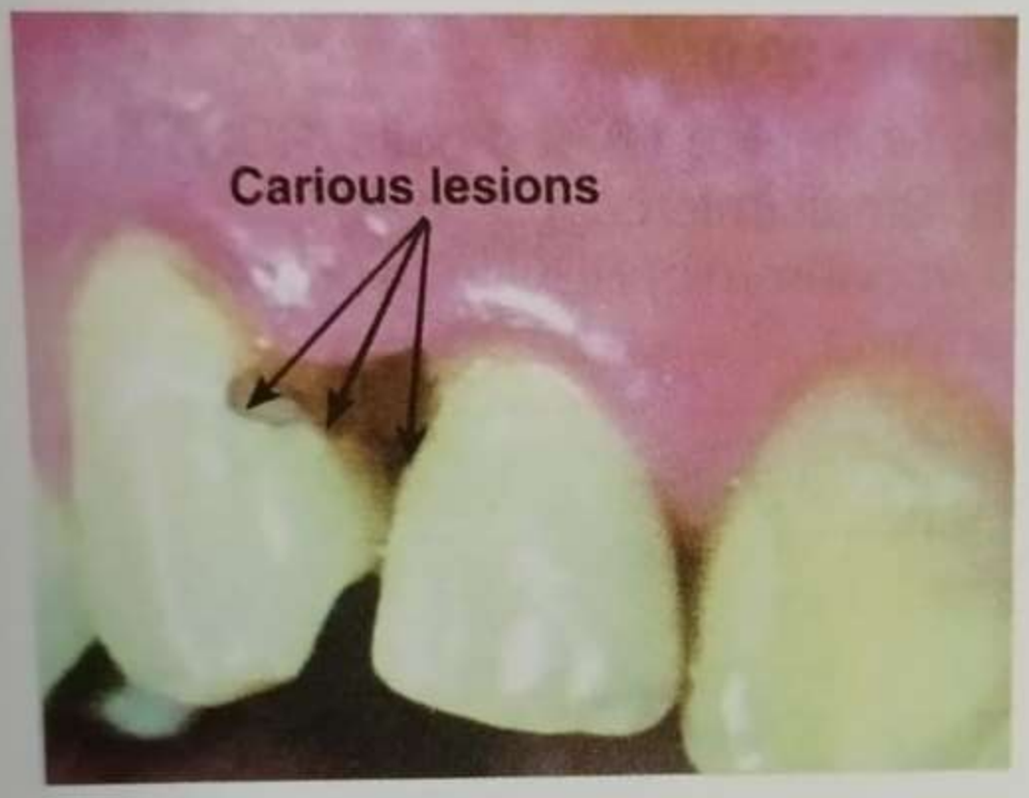



**Figure 29.12**
Dental caries (cavities).

### E. Other micromine

Chromium (Cr) and
ates the action of
fruits, vegetables,
added to water in
dental caries (Fig.
ylapatite, forming
dissolving acid pro

## IV. ULTRATRACE

The ultratrace minerals
num (Mo). They are req

### A. Iodine

Iodine is utilized in th
nine ($T_3$) and thyroxin
and metabolism. Cir
concentrated in the
then is sent into the co



collagen fibrils →



**Figure 4.8**
The legs of a 46-year-old man with scurvy.



**Figure 21.6**
Skin eruptions in a patient with porphyria cutanea tarda.



**Figure 21.7**
Urine from a patient with porphyria cutanea tarda (**right**) and from a patient with normal porphyrin excretion (**left**).

a. **Chronic hepatic porphyria:** PCT, the most common porphyria, is a chronic disease of the liver that is associated with severe deficiency of UROD. Although the cause of the deficiency is influenced by various factors, including hepatic iron overload, exposure to halogenated hydrocarbons, estrogen therapy, and the presence of hepatitis C or HIV infections. (Note: Mutations to the UROD gene occur in 20% of affected individuals. Inheritance is autosomal dominant.) Onset is typically during the fourth or fifth decade. Excessive porphyrin accumulation leads to cutaneous lesions (Fig. 21.7) as well as urine that is red to brown in natural light and pink to red in fluorescent light.

b. **Acute hepatic porphyrias:** Acute intermittent porphyria, ALA dehydratase–deficiency porphyria, and hereditary coproporphyria are characterized by acute attacks of gastrointestinal, neurologic, psychiatric, and motor symptoms that may be accompanied by photosensitivity (Fig. 21.8). Porphyria is caused by overproduction of ALA and PBG, such as AIP and neuropsychiatric disturbances, which may progress to delirium. Symptoms of the acute hepatic porphyrias may be precipitated by use of drugs, such as barbiturates and ethanol, which induce the synthesis of the rate-limiting enzyme of the microsomal drug-oxidation system. This depletes the amount of available heme, which, in turn, increases the synthesis of ALAS1.

c. **Erythropoietic porphyrias:** The congenital erythropoietic porphyrias (CEP and EPP) cause photosensitivity characterized by skin rashes and blisters that appear in childhood (Fig. 21.8).

2. **Increased δ-aminolevulinic acid synthase activity** is a common feature of the hepatic porphyrias. ALAS1 is the rate-limiting enzyme in the synthesis of heme. In the liver, heme normally functions to repress the ALAS1 gene. Therefore, the absence of heme (or hemin) causes an increase in the synthesis of ALAS1 and the porphyrin precursors. This causes an increased synthesis of the porphyrin precursors prior to the genetic block. The accumulation of these precursors is the major pathophysiology of the porphyria attacks.

**Figure 29.16**
Exophthalmos.

| Mo-REQUIRING ENZYME | FUNCTION |
|---|---|
| Aldehyde oxidase | Metabolizes drugs |
| Sulfite oxidase | Converts sulfite to sulfate in metabolism of the sulfur-containing amino acids methionine and cysteine |
| Xanthine oxidase | Oxidizes hypoxanthine to xanthine and xanthine to uric acid in purine degradation |

**Figure 29.17**
Enzymes (oxidases) that require molybdenum (Mo).

Cutaneou
adults).

## C. Molybde

Molybden
mammalia
sources. M
toxicity in



Co
$B_{12}$
bal
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p. 2
Refe

# V. CHAPTER

The minerals are

**Figure 18.12**
Enterohepatic circulation of bile salts. (Note: Ionized bile



**Figure 18.13**
Gallbladder with gallstones.

Choleslithiases occurs due to
1) ↓ bile acid in bile-
2) ↑ cholestrol secretion in bile-

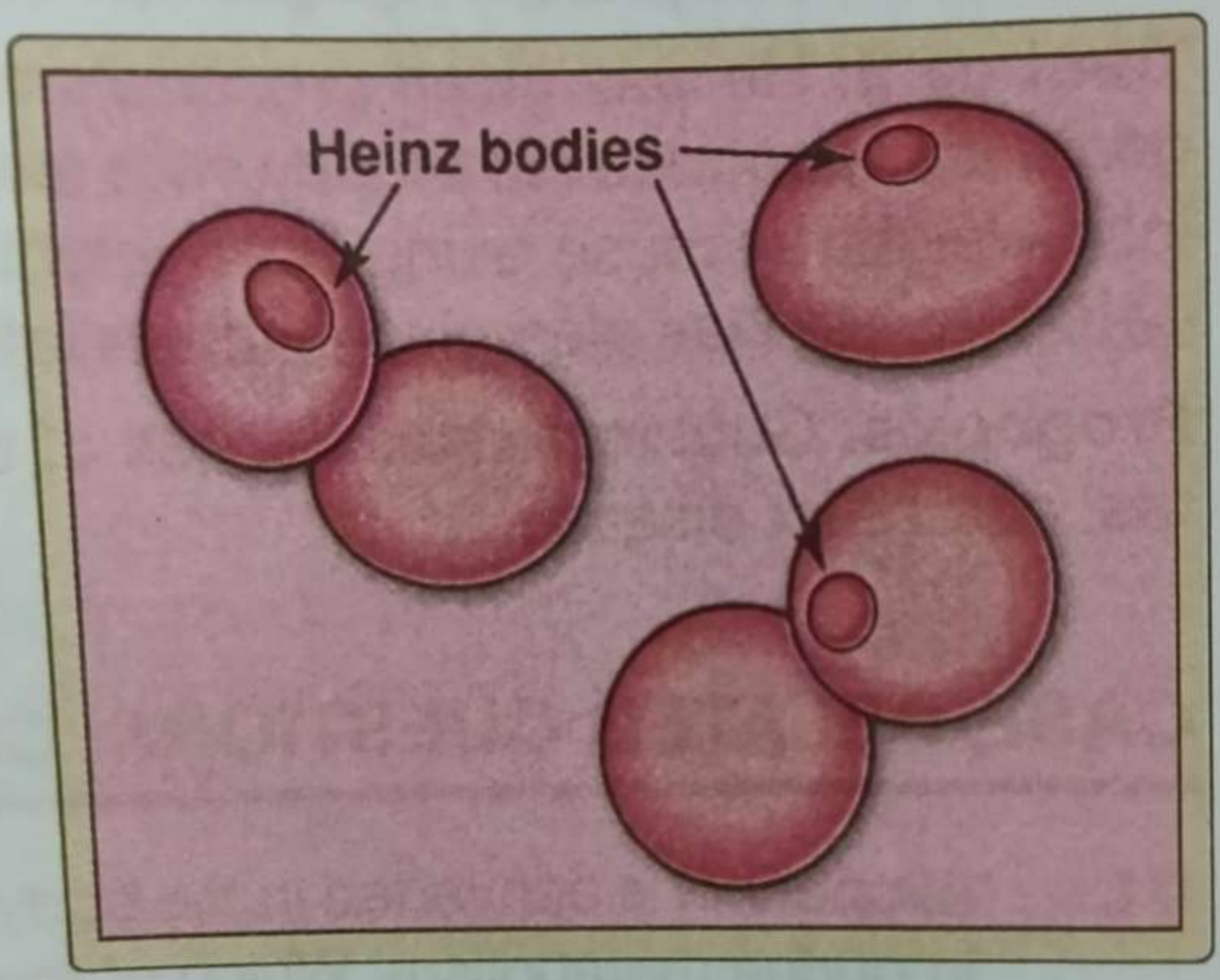
F. **Bile sa**

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stones
mixed

## VI. PLA

The plasm
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de antibiotic and a urinary analgesic for a
olor (become reddish) with the analgesic,
days. Last night, his wife noticed that his




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C

n affected individuals exposed to oxidative
ill be switched to a different antibiotic and
to medical providers. He likely was not pre-
e had this genetic defect.

6PD deficiency does not cause mortality or

**t answer**

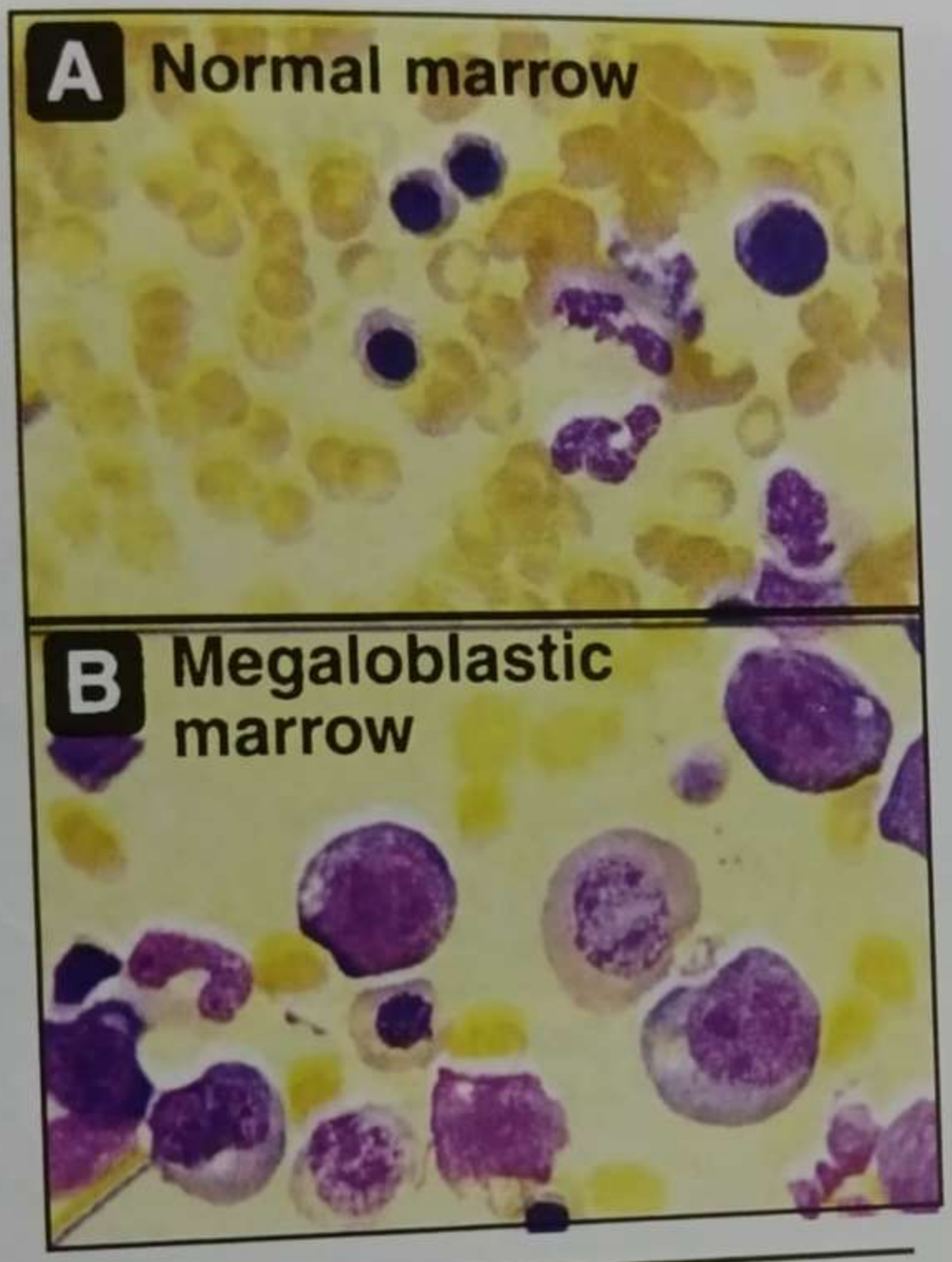
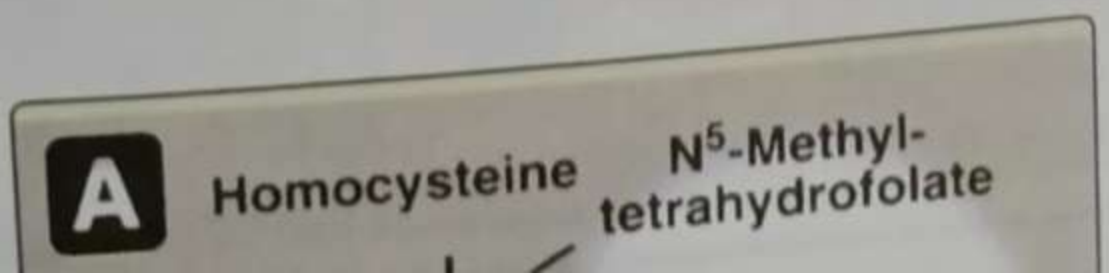



**Figure 28.4**
Bone marrow histology in normal (**A**) and folate-deficient (**B**) individuals.



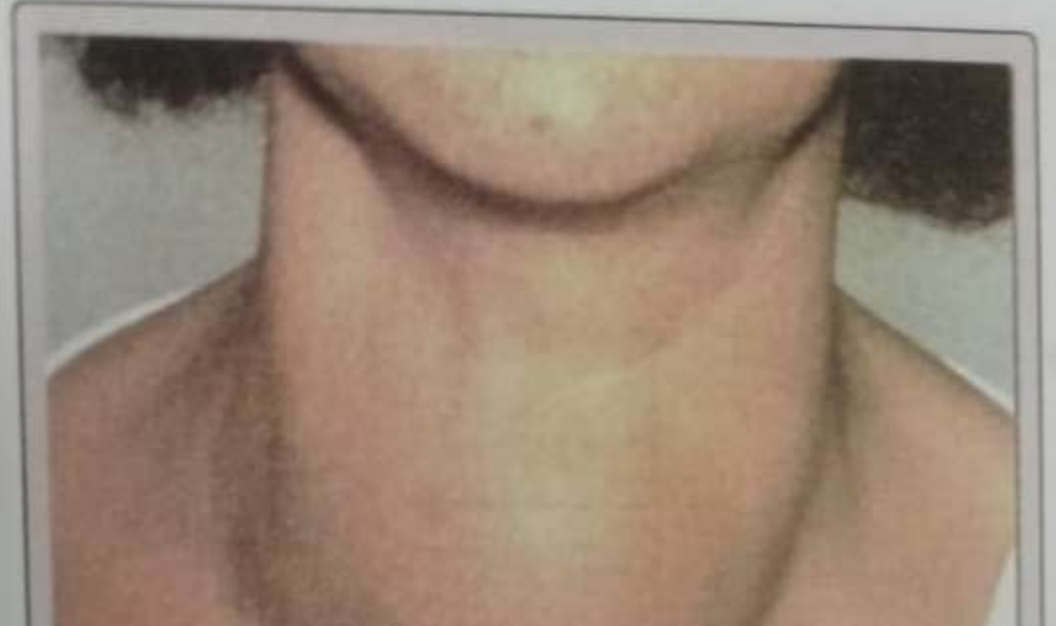


**Figure 29.15**
Goiter.



**Figure 29.16**
Exophthalmos.

| Mo-REQUIRING ENZYME | FUNCTION |
|---|---|
| Aldehyde oxidase | Metabolizes drugs |

destruction of TPO
hypothyroidism].)

2. **Hyperthyroidism:**
of thyroid hormone
of I-containing sup
common cause of
an antibody that m
in dysregulated pr
nervousness, weig
protruding eyes (ex

## B. Selenium

Selenium (Se) is presen
proteins) as a constitue
derived from serine (se
**peroxidase** that oxidize
oxide, a ROS, to water
reduces thioredoxin, a
p. 330); and **deiodinase**
Meat, dairy products, a
han disease, first ident
by eating foods produc
caused by overingestion
Cutaneous and neurolog
adults).

## C. Molybdenum

Molybdenum (Mo) funct
mammalian **oxidases** (F
sources. No dietary defi
toxicity in humans (UL =

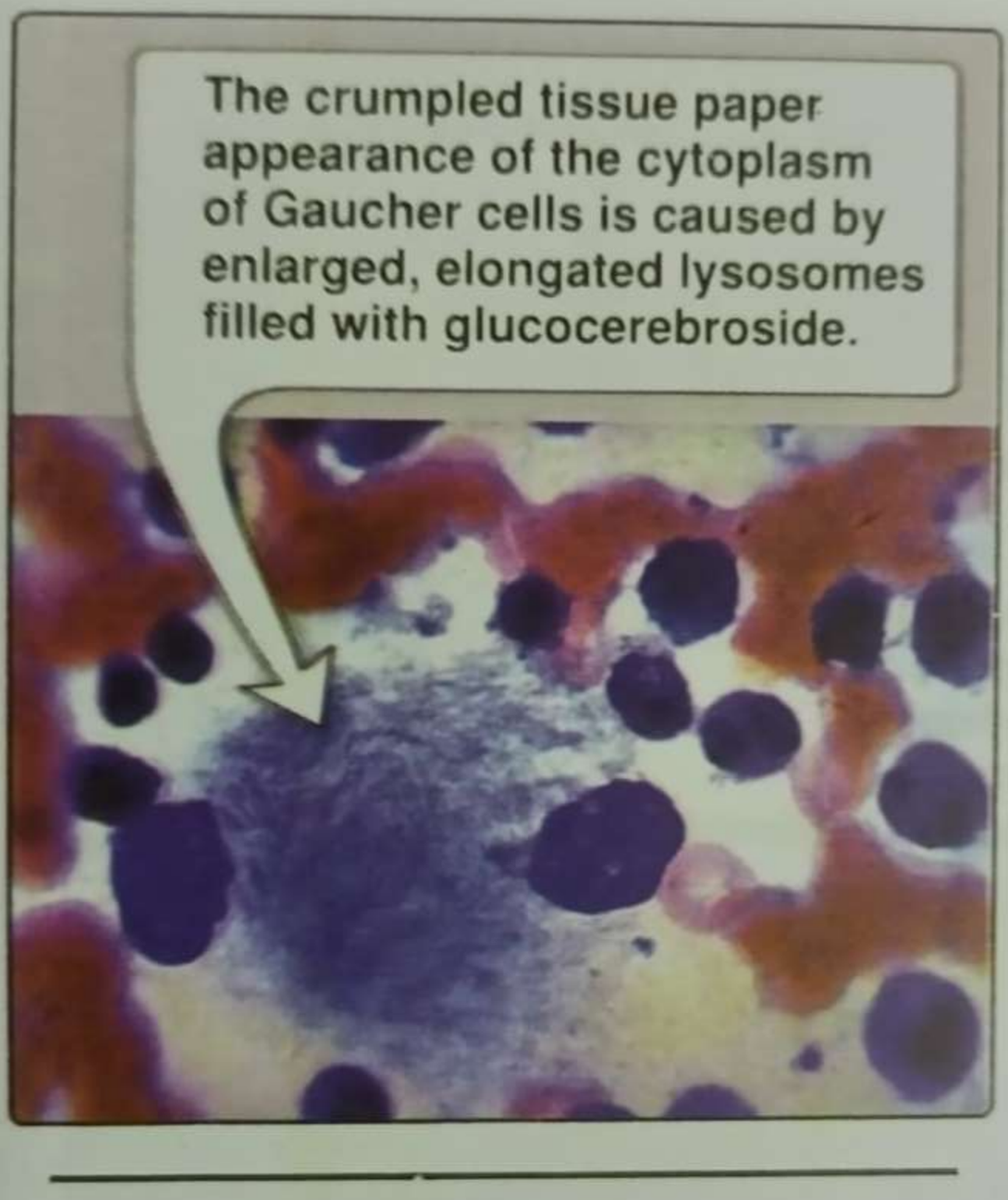
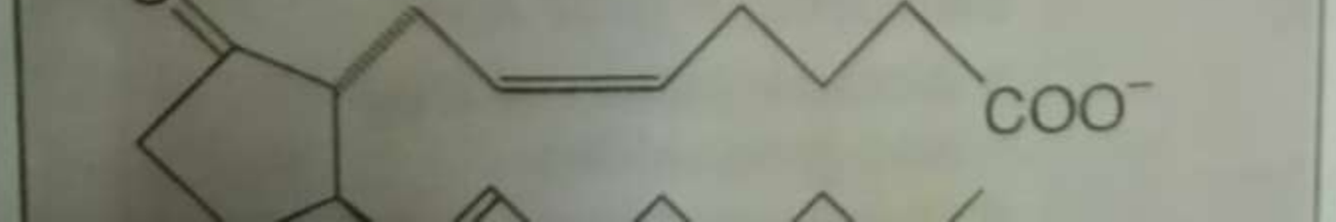



**Figure 17.20**
Aspirated bone marrow cells from a patient with Gaucher disease.




## VIII. E
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cally resu
Examples

### A. Prosta

the mutase requires vitamin $B_{12}$, the disease can be severe $B_{12}$ deficiency. The breakdown of [illegible] acids, valine, isoleucine, methionine, and threonine MMA, due to this enzyme deficiency. Elevation of the blood and urine can result in a metabolic acidosis be an increase in propionyl-CoA, exacerbating the accumulation of additional propionic acid. Symptoms infancy, varying due to the degree of the enzyme ing failure to thrive, vomiting, dehydration, hypotonia delay, seizures, hepatomegaly, hyperammonemia, encephalopathy. If severe and left untreated, it can disability, chronic renal or hepatic damage, coma or death. Treatment includes a low-protein, high- vitamin $B_{12}$ supplementation. The diet limits the threonine, methionine, and valine, as these amino the buildup of methylmalonic acid by the mutase

## VII. Chapter Summary

- **Amino acids** whose catabolism yields pyruvate or [illegible] the **TCA cycle** are termed **glucogenic** (Fig. 20.24). to the net formation of **glucose** in the **liver** and **kidney**. glucogenic amino acids are glutamine, glutamate, pro- tidine, alanine, serine, glycine, cysteine, methionine, aspartate, and asparagine.
- Amino acids whose catabolism yields either acetyl CoA out pyruvate serving as an intermediate) or acetoacetate sor acetoacetyl CoA) are termed **ketogenic**. Leucine is solely ketogenic.
- Tyrosine, phenylalanine, tryptophan, and isoleucine are and glucogenic.
- **Nonessential amino acids** can be synthesized from mediates or from the carbon skeletons of essential ami
- **Essential amino acids** need to be obtained from the di histidine, methionine, threonine, valine, isoleucine, phe tophan, leucine, and lysine.
- **PKU** is caused by a **deficiency** of **PAH**, which converts to tyrosine. **Hyperphenylalaninemia** may also be caused cies in the enzymes that synthesize or regenerate the co PAH, $BH_4$. Untreated individuals with PKU suffer from disability, developmental delay, microcephaly, seizures, teristic musty (mousy) smell of the urine. Treatment is dietary phenylalanine. **Tyrosine** becomes an essential nent for people with PKU.
- **MSUD** is caused by a partial or complete deficiency of enzyme that decarboxylates the **BCAAs, leucine, isoleu valine**. Symptoms include feeding problems, vomiting, changes in muscle tone, and a characteristic sweet smell untreated, the disease leads to neurologic problems, co Treatment involves controlling BCAA intake.
- Other important genetic diseases associated with amino lism include **albinism**, **homocystinuria**, **MMA**, **alkapto- nemia**, **tyrosinemia**, and **cystathioninuria**.

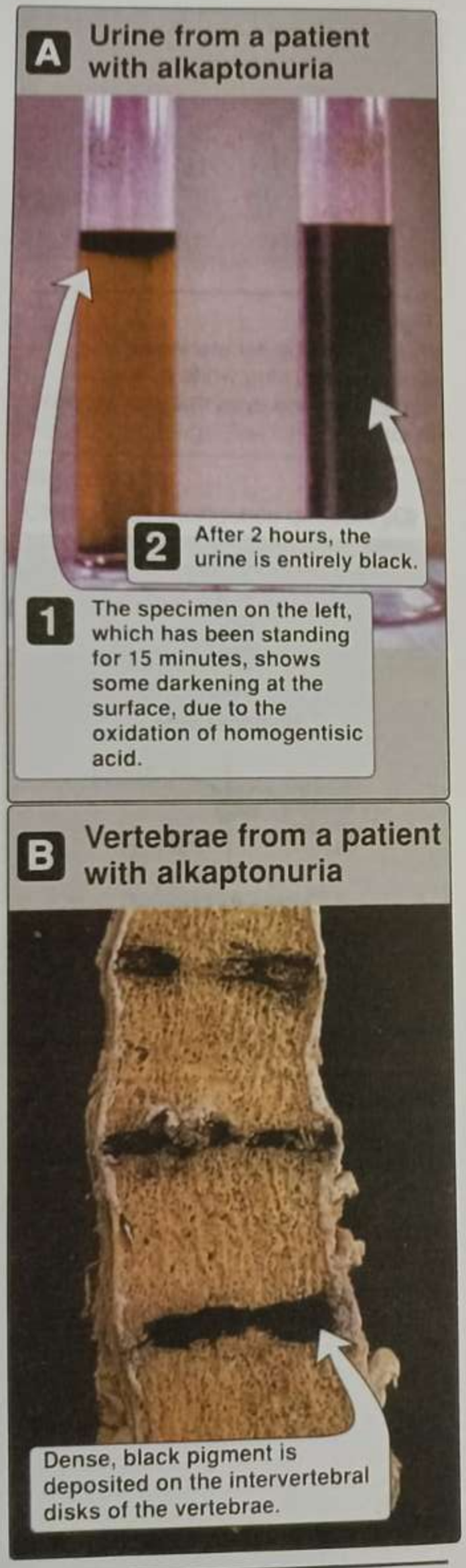



**Figure 20.23**
Specimens from a patient with alkaptonuria. **A:** Urine. **B:** Vertebrae.



Metabolism

Catabolism

Removal of α-amino group

Synthe

involves

Figure 20.24
Key concept m



**Figure 20.21**
Patient with oculocutaneous albinism, showing white eyebrows and lashes and eyes that appear red in color.

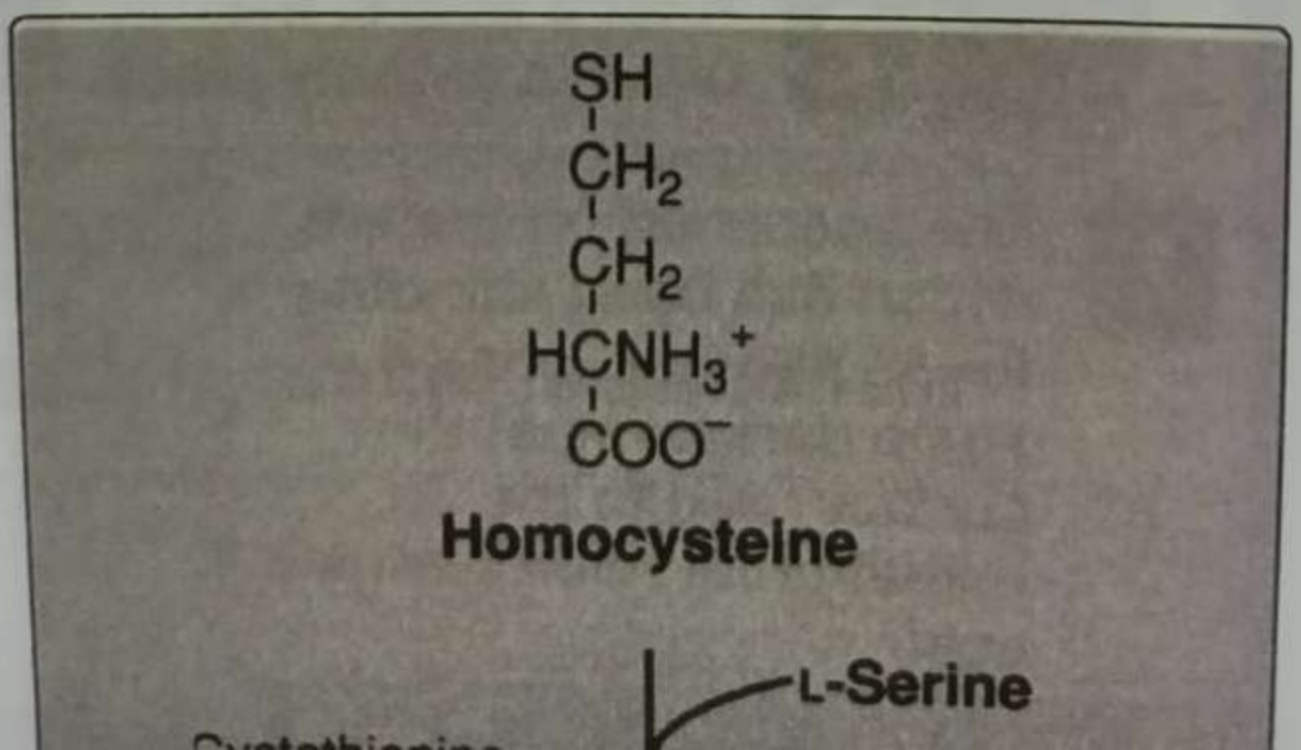




**Figure 22.16**
Tophaceous gout.

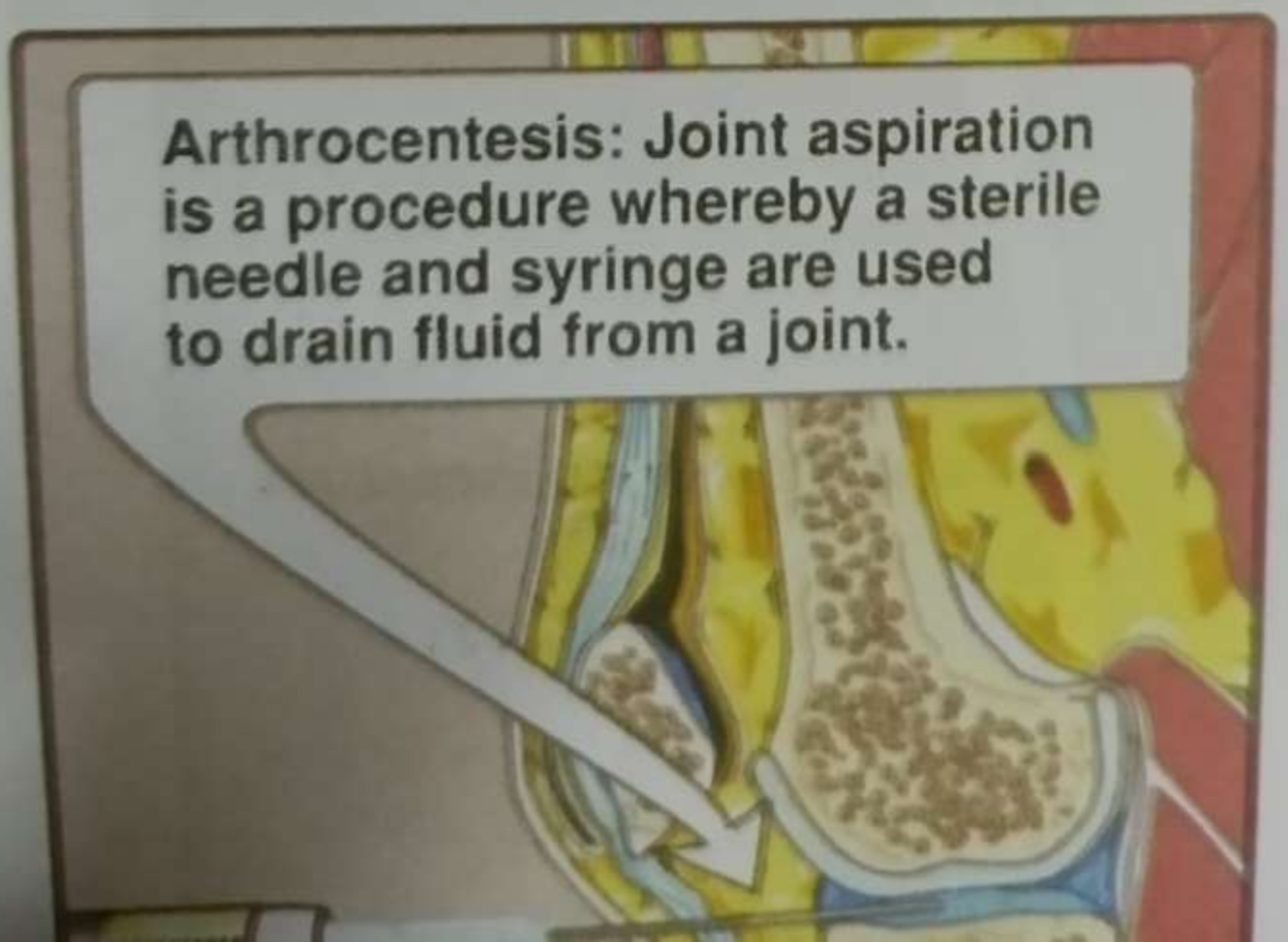




**Figure 28.25**
Bowed legs of middle-aged man with osteomalacia, a nutritional vitamin D deficiency that results in demineralization of the skeleton.

3. **Hypoparathyroidism:** La hyperphosphatemia. (Note Patients may be treated wit

### F. Toxicity

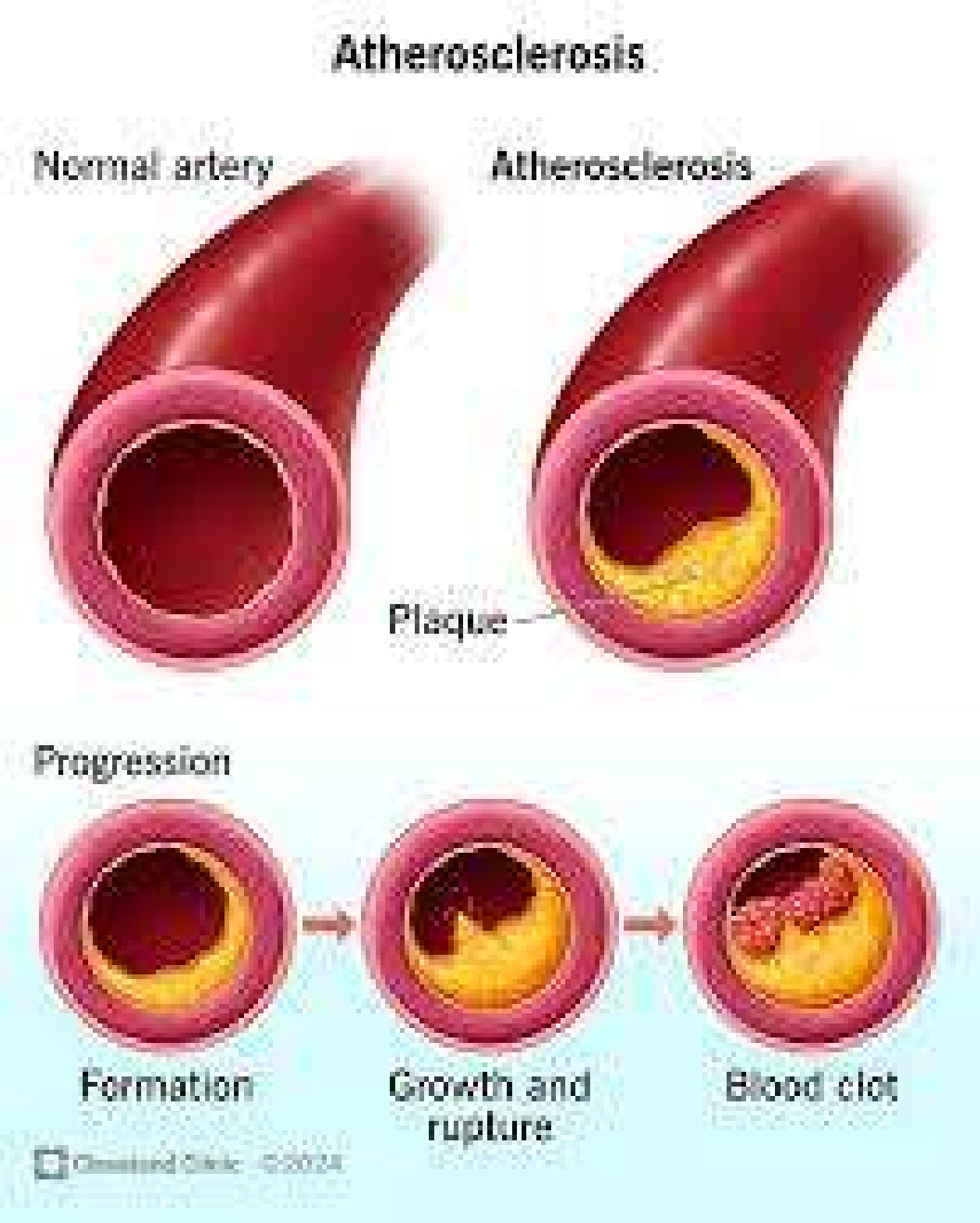
Like all fat-soluble vitamins and is only slowly metaboliz months) can cause loss of Enhanced $Ca^{2+}$ absorption cemia, which can lead to d (metastatic calcification). Th individuals ages 9 years or age 9 years. (Note: Toxicity Excess vitamin D produced

## XIII. VITAMIN K

The principal role of vitamin K a number of proteins (most of which it serves as a coenzyme acid residues in these proteins for example, in plants as phyll bacteria as menaquinone (or v menadione, is able to be conv

### A. Function

1. **γ-Carboxyglutamate** posttranslational modifi FVII, FIX and FX (See


Atherosclerosis
Normal artery
Atherosclerosis
Plaque
Progression
Formation
Growth and rupture
Blood clot

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**Figure 26.5**
**A:** Patient with leptin deficiency before initiation of therapy at age 5 years. **B:** Patient at age 9 years after 48 months of therapy with subcutaneous injections of recombinant leptin.

type B is the nonneuropathic form. It has a later age of onset and a longer survival time than type A.)

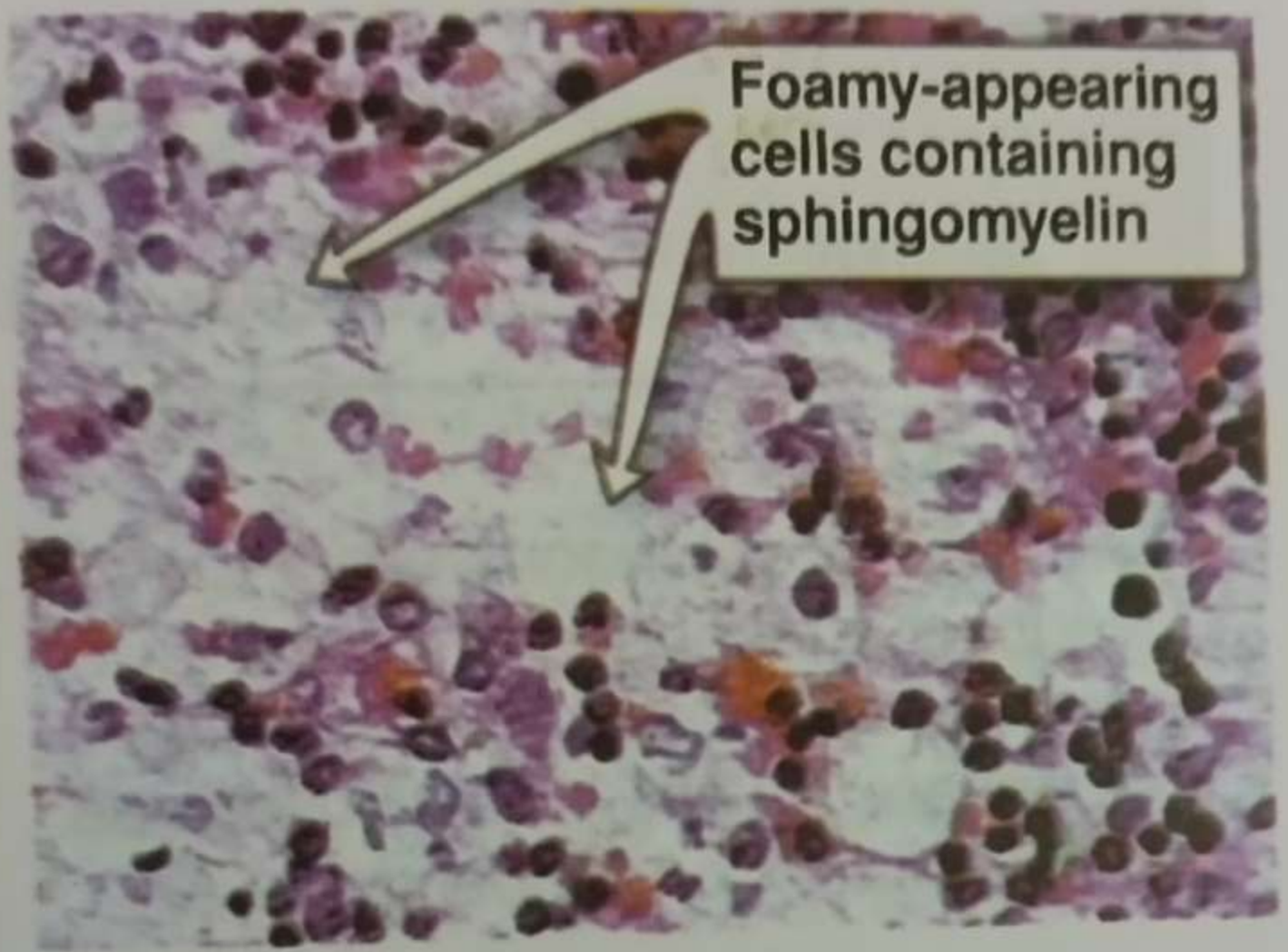
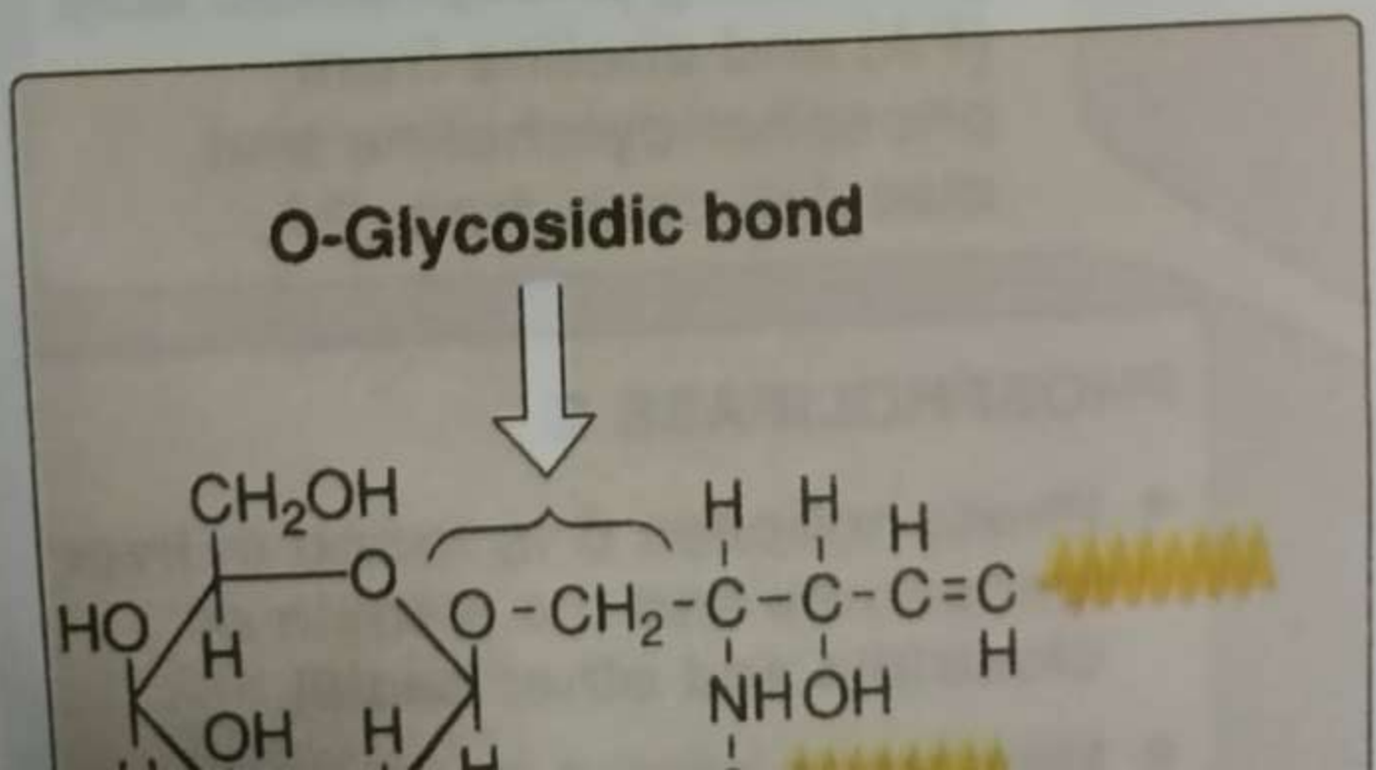



**Figure 17.13**
Accumulation of lipids in spleen cells from a patient with Niemann–Pick disease.




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**B. Acid**

Acid
pH
([NA
sulf

1.



**Figure 22.18**

Gout can be diagnosed by the presence of negatively birefringent monosodium urate crystals in aspirated synovial fluid examined by polarized light microscopy. Here, crystals are seen within polymorphonuclear leukocytes.

rate are elevated. Lipid deposits are noted on the periphery of his corneas (cornea
d under the skin on and around his eyelids (xanthelasmas; *see right image*). No depos
omas) are detected.

**Corneal arcus**



**Xanthelasmas**



**s:** The patient's electrocardiogram is consistent with an acute myocardial infarction (MI). areas of severe stenosis (narrowing) of several coronary arteries. Initial results from the ude the following:

| | **Patient** | **Reference Range** |
|---|---|---|
| | 0.5 ng/mL | Below 0.04 ng/mL |
| rotein (LDL) cholesterol | 365 mg/dL (**H**) | |
| protein (HDL) cholesterol | 304 mg/dL (**H**) | <200 |
| cylglycerols) | | |

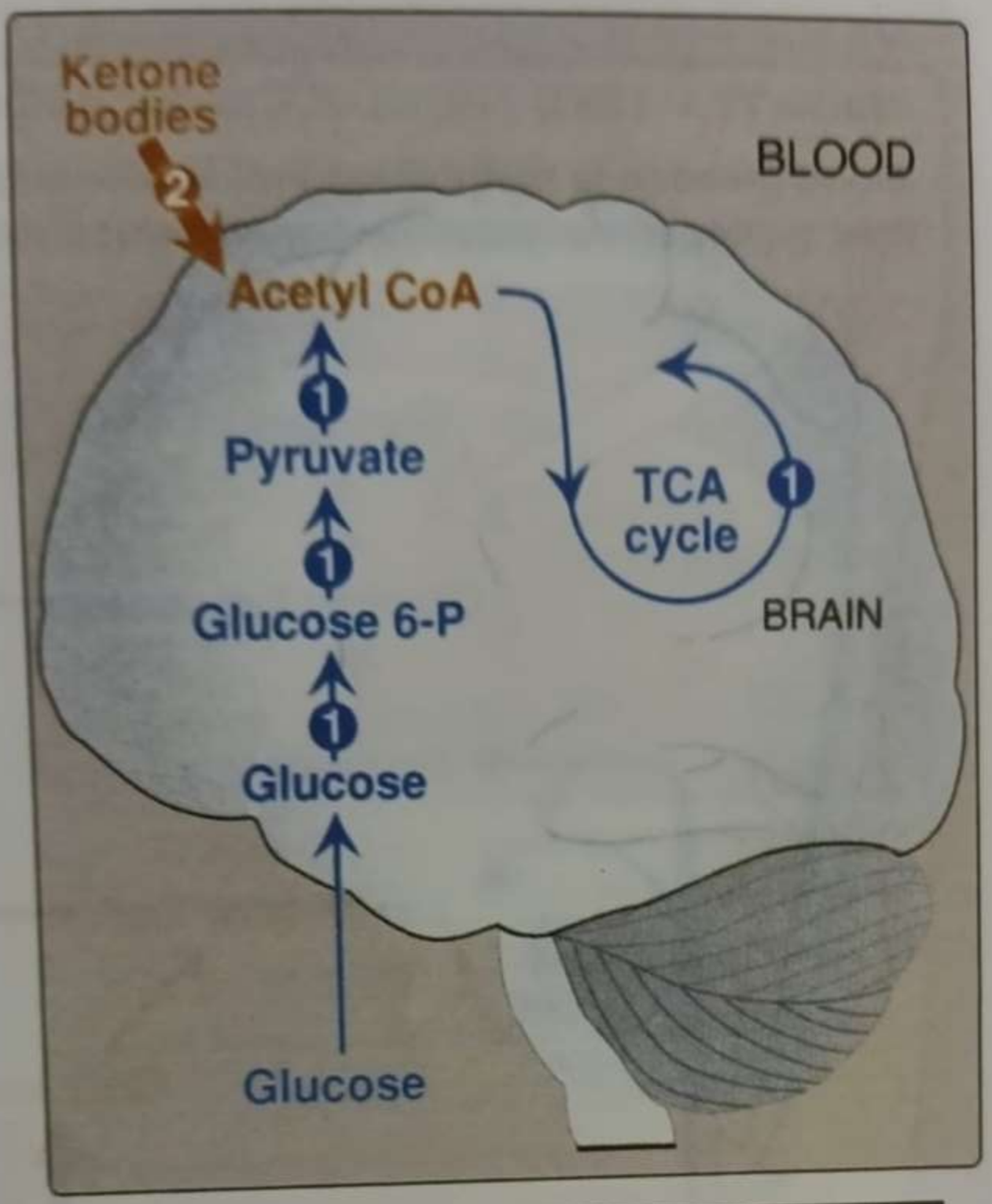



**Figure 24.16**
Major metabolic pathways in the brain during fasting. (Note: The *numbers in the circles*, which appear both in the figure and in the corresponding citation in the text, indicate important pathways for metabolism of fat or carbohydrates.) CoA = coenzyme A; TCA = tricarboxylic acid; P = phosphate.

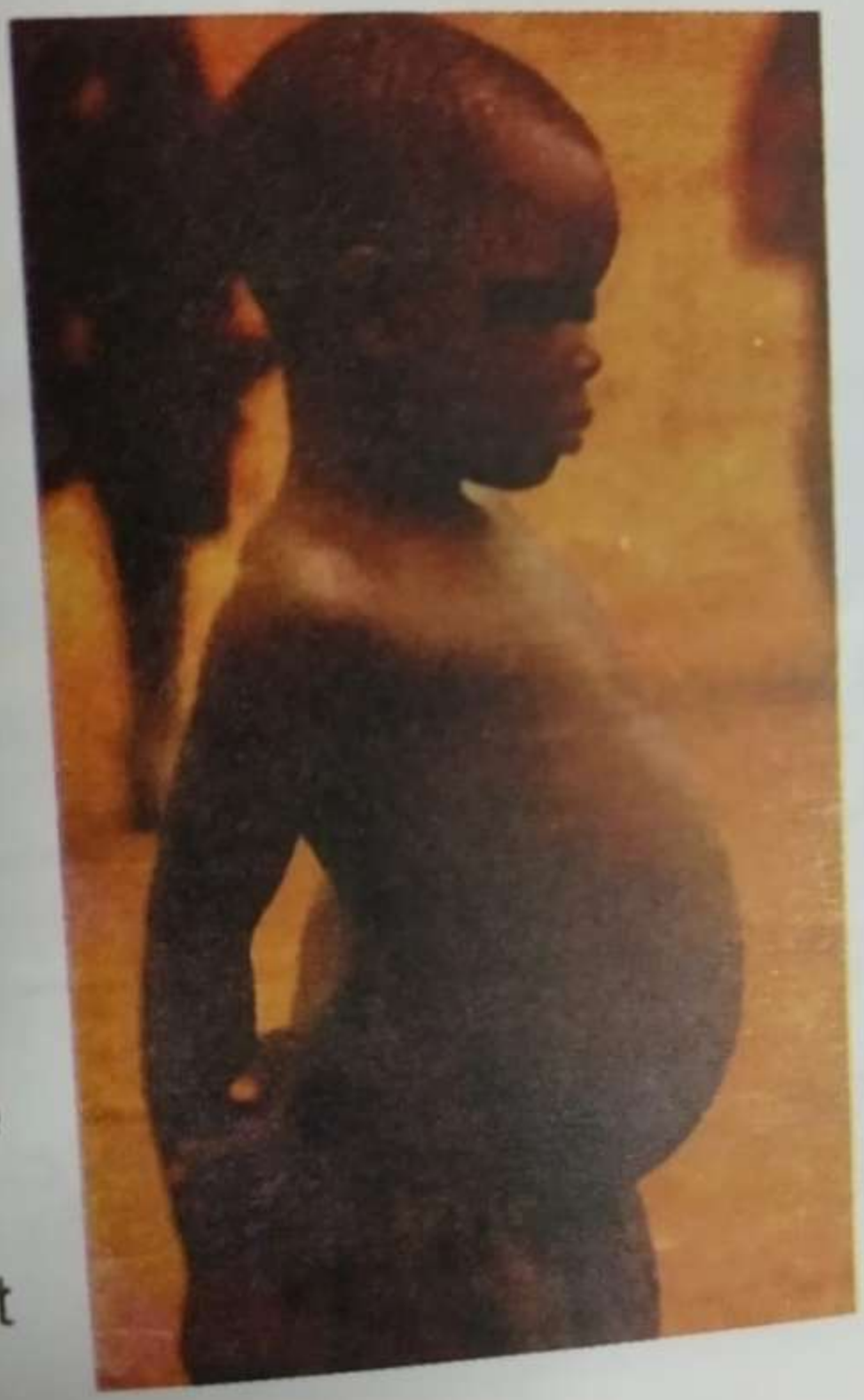
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The correct answer = B. K
inadequate protein intake
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patient with kwashiorkor
peripheral edema (note the
caused largely by a decrea
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**Figure 22.11**
Lesions on the lips of a patient with Lesch–Nyhan syndrome.

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## IV. DEOXYRIBON

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### A. Ribonucleotide re

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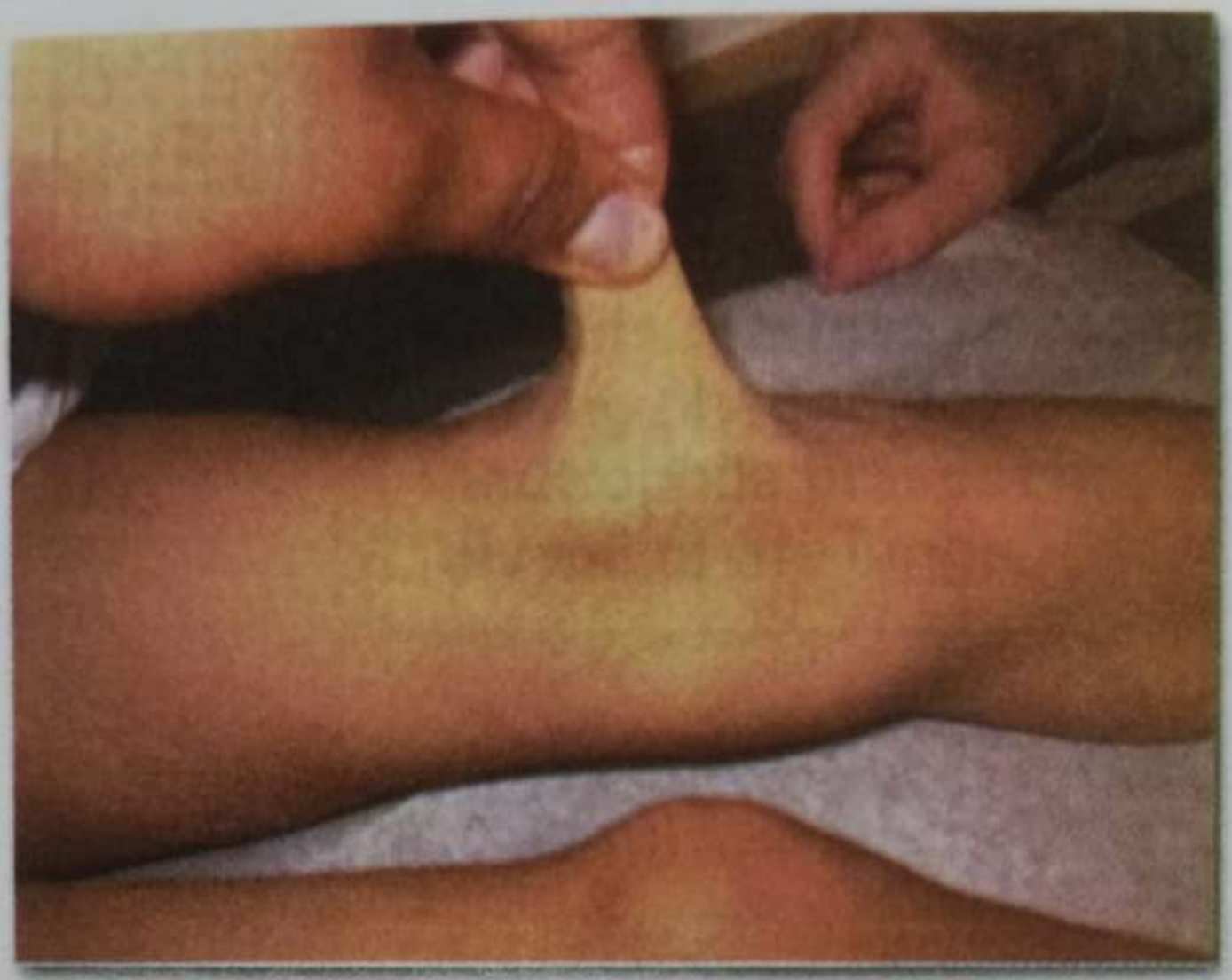


**Figure 4.10**
Stretchy skin of classic Ehlers–Danlos syndrome and mechanism of bisphosphonates.

### C. Secretion regulation

Secretion of insulin is regulated by blood-borne fuels and

1. **Increased secretion:** Insulin secretion by the
cells is closely coordinated with the secretion of

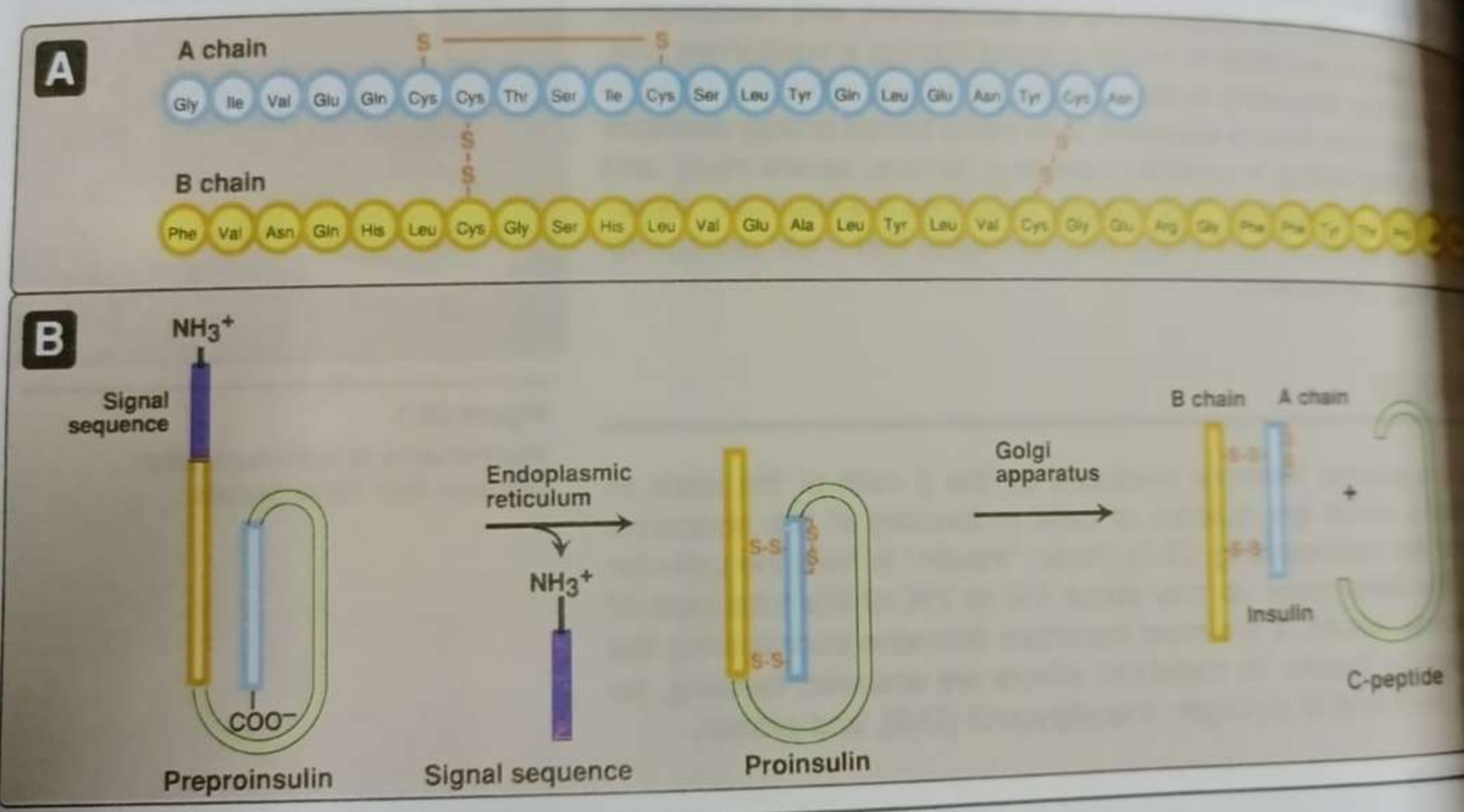



**Figure 23.3**
**A:** Structure of insulin. **B:** Formation of human insulin from preproinsulin. S-S = disulfide bond.

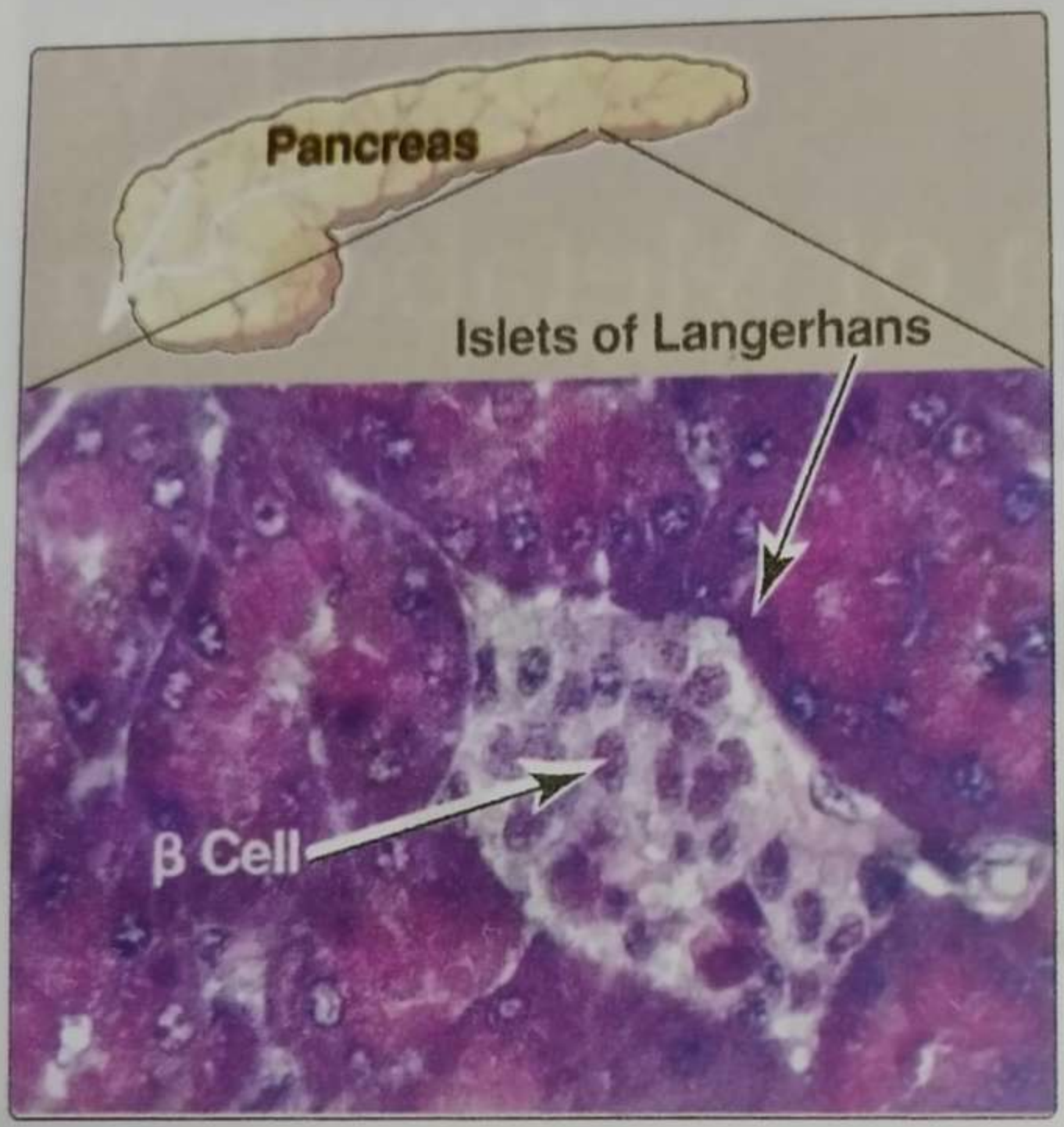



**Figure 23.2**
Islets of Langerhans.

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## B. Synthesis

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## C. Secretion

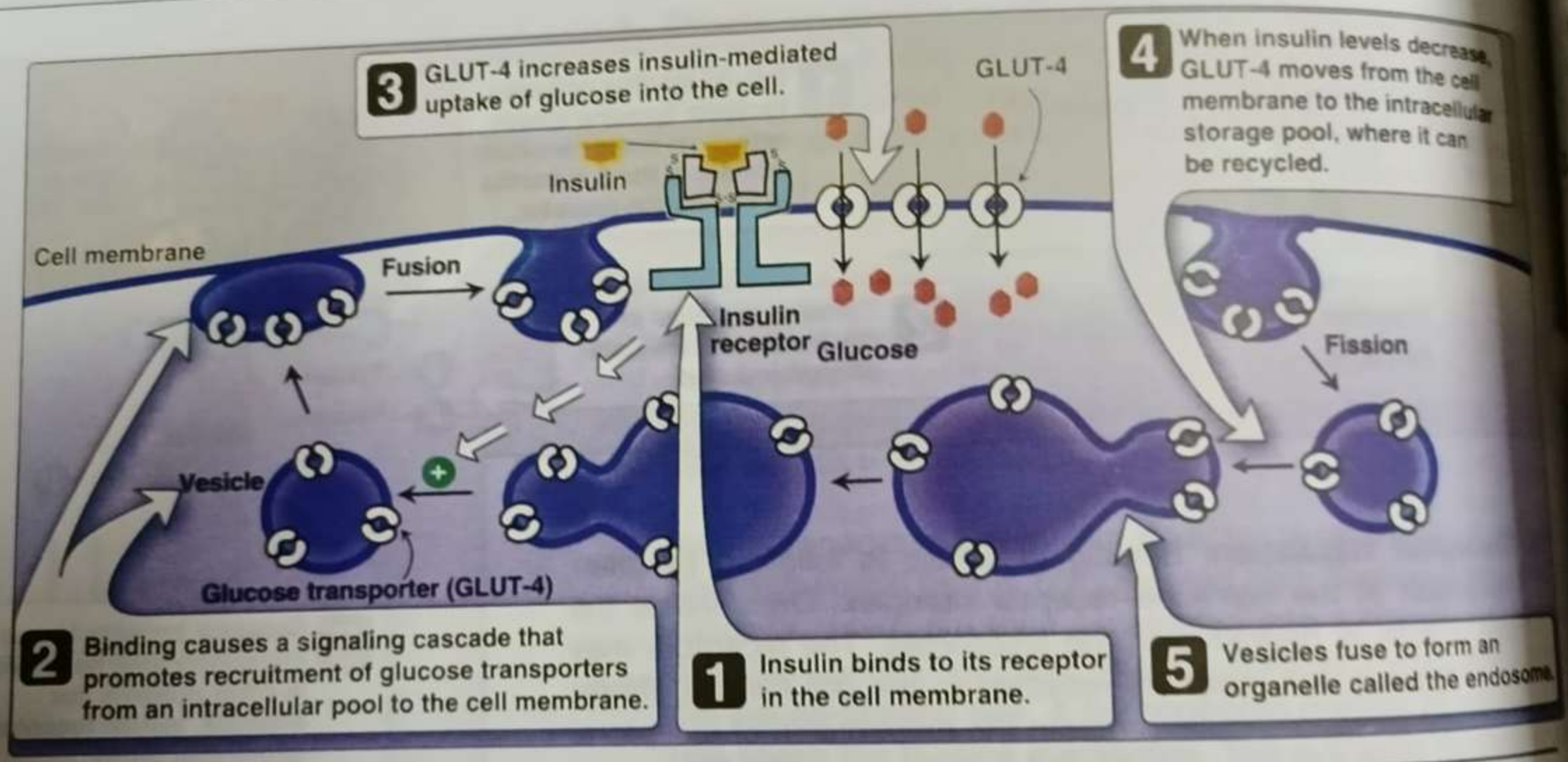



**Figure 23.8**
Insulin-mediated recruitment of GLUT-4 from intracellular stores to the cell membrane in skeletal and cardiac muscle and adipose tissue. S-S = disulfide bond.

| Active | Facilitated |
|---|---|

### A. Increased secretion




**Figure 27.19**
**A:** A child with kwashiorkor. Note the swollen belly and lower legs. **B:** Child suffering with marasmus.

1. **Kwashiorkor:** Kwashiorkor occurs when pr
is relatively greater than the reduction in tota
deprivation is associated with severely decre
visceral protein. Kwashiorkor is commonly se
countries in children after weaning at about age
diet consists predominantly of carbohydrates.
include stunted growth, skin lesions, depigmen
fatty liver, bilateral pitting edema, and decrease
concentration. Edema results from the lack o
proteins, primarily albumin, to maintain the dis
between blood and tissues. It may mask mus
Therefore, chronic malnutrition is reflected in t
albumin. (Note: Because caloric intake from ca
be adequate, insulin levels suppress lipolysis
Kwashiorkor is, therefore, nonadapted malnutri

2. **Marasmus:** Marasmus occurs when calorie de
tively greater than the reduction in protein. It u
developing countries in children younger than a
breast milk is supplemented or replaced with
native cereals that are usually deficient in both p
ries. Typical symptoms include arrested growth,
wasting and depletion of subcutaneous fat (em
ness, and anemia (Fig. 27.19). Individuals wit
not show the edema observed in kwashiorkor.
severely malnourished individuals can result in
mia (see Chapter 29 Section II A2), because any
phate is used to phosphorylate carbohydrate inte
is frequently given because it is rich in phosphate

Cachexia, a wasting disorder characterized by los
tite and muscle atrophy (with or without increase
that cannot be reversed by conventional nutritiona
is seen with a number of chronic diseases, such
and chronic pulmonary and renal disease. It is assoc
decreased treatment tolerance and response and
survival time.

## VIII. NUTRITION TOOLS

A set of tools has been developed that gives consumers
about what (and how much) they should eat as well as
content of the foods they do eat. Additional tools allow me
signals to assess whether or not the nutritional needs of



**Figure 28.9**
Oral manifestations in a patient with scurvy.

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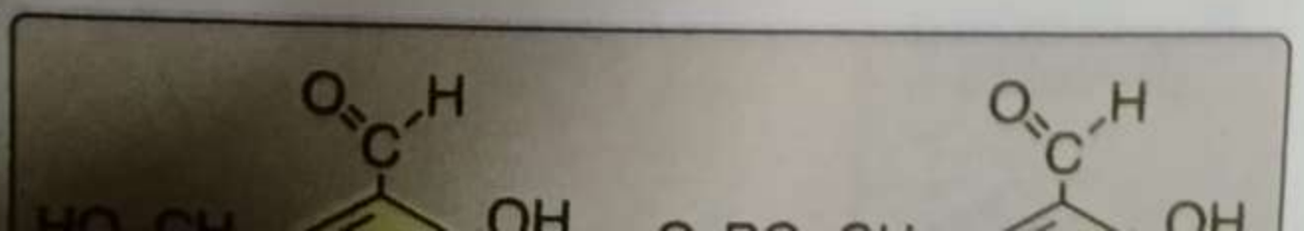
**A. Defic**

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**B. Chron**

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## V. PYRID



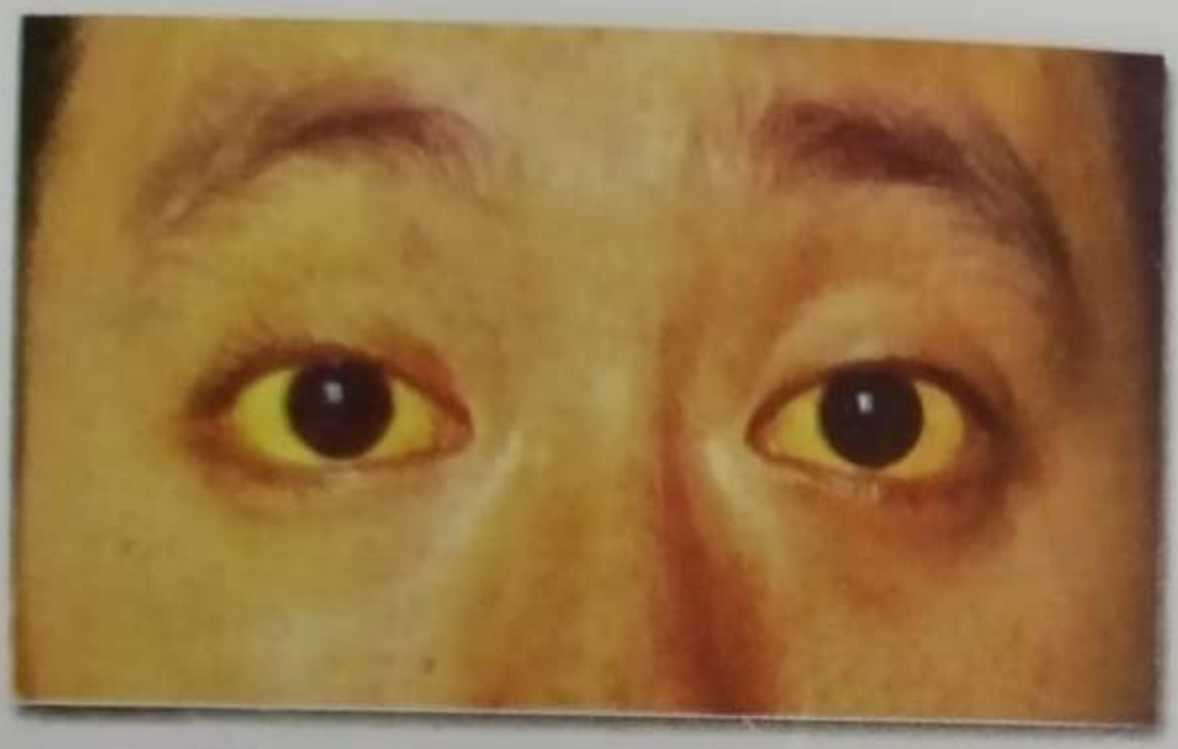
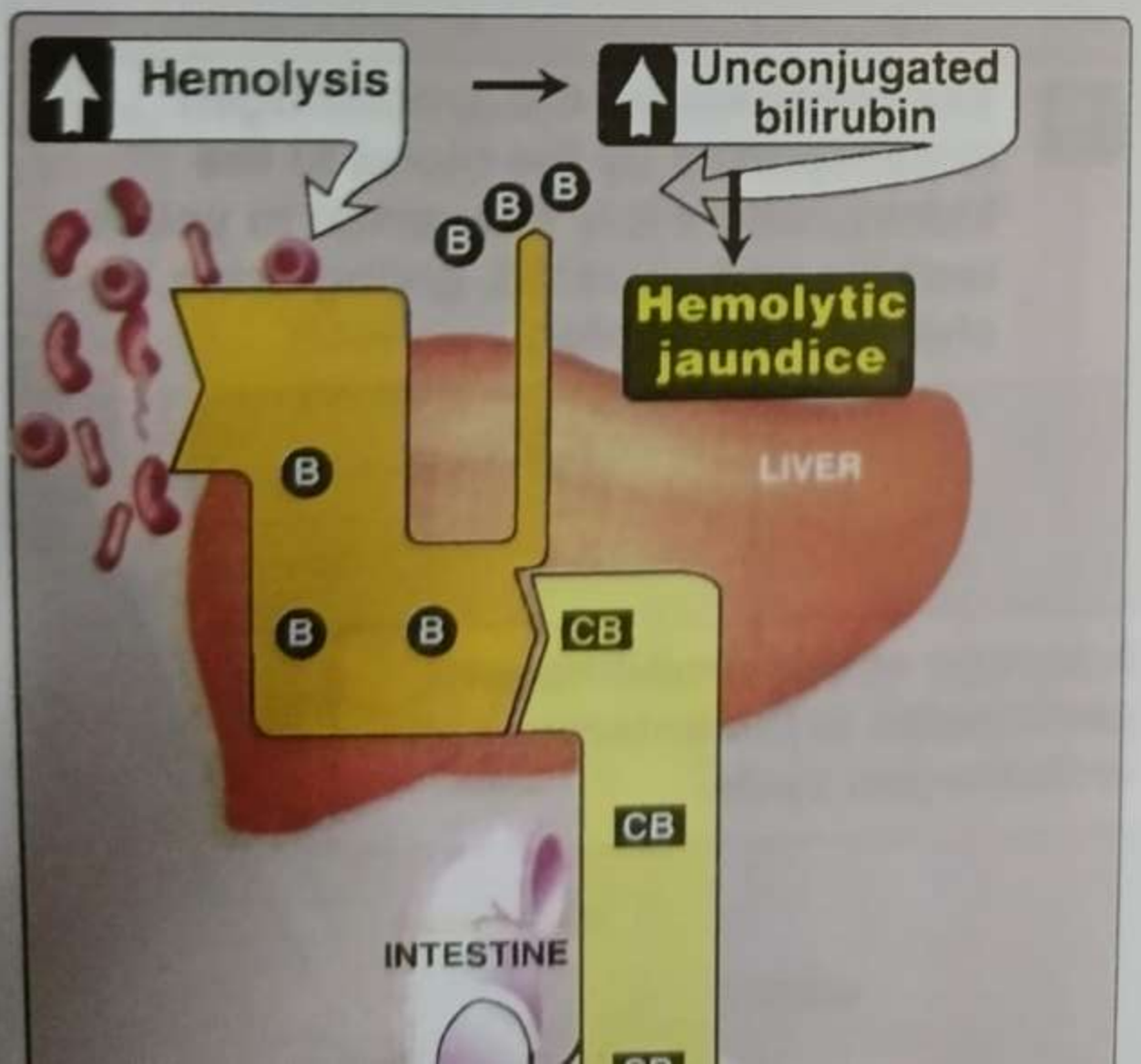


**Figure 21.11**
Jaundiced patient with the sclerae of his eyes appearing yellow.




## E. Jaundice

Jaundice (or, icter
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shown in Figure 2
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1. **Types:** Jaundi
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